

DAY RESEARCH

Smartwatches

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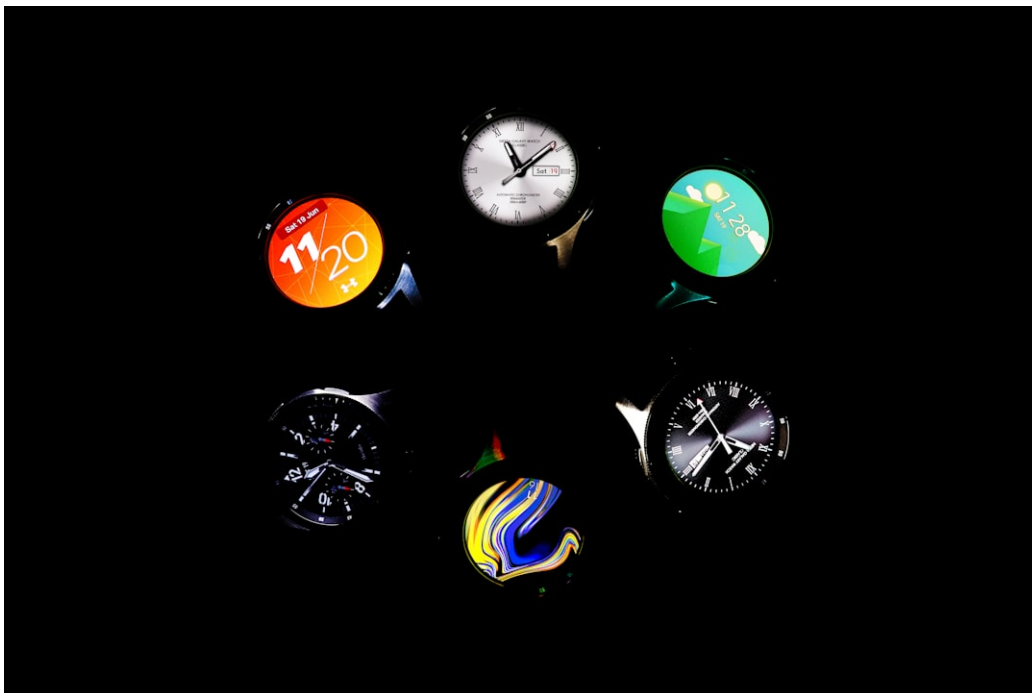


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Executive digest



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Smartwatch Research Digest

Introduction to Smartwatches

The topic of smartwatches has been explored, with a focus on their battery life and display technologies. The research highlights the importance of considering the specific needs of the user when selecting a smartwatch.

Battery Life and Performance

A key finding is that the best battery life smartwatches may not always perform as expected, with actual performance varying from advertised claims. This discrepancy emphasizes the need for thorough research and careful consideration of user needs.

Display Technologies

MIP (Memory-in-Pixel) displays, utilized by brands such as Garmin and COROS, have been identified as a power-efficient technology. These displays draw power only when the pixels change, contributing to extended battery life. This technology is a notable aspect of smartwatch design, allowing for a balance between feature-richness and power efficiency.

Balancing Smart Features and Battery Life

The research underscores the importance of finding the perfect balance between smart features and lasting power. To achieve this balance, it is crucial to research thoroughly and consider the specific needs of the user. This may involve evaluating the importance of various features, such as fitness tracking, notification support, and music storage, against the need for extended battery life.

Key Findings and Takeaways

- The actual battery life of smartwatches may vary from advertised claims.
- MIP displays are a power-efficient technology, drawing power only when pixels change.
- Thorough research and consideration of user needs are essential for finding the perfect balance between smart features and battery life.

Open Questions and Contradictions

While the research highlights the importance of balancing smart features and battery life, it does not provide a clear framework for evaluating these trade-offs. Further exploration is needed to develop a comprehensive understanding of user needs and preferences.

Practical Takeaways

- When selecting a smartwatch, prioritize features that align with your specific needs.
- Consider the display technology used, as power-efficient options like MIP displays can contribute to extended battery life.
- Be aware that actual battery life may vary from advertised claims, and plan accordingly.

Next Angles to Explore

- Investigating user preferences and needs to develop a framework for evaluating smartwatch trade-offs.
- Examining the impact of different display technologies on battery life and overall user experience.
- Analyzing the role of software optimization in enhancing battery life and performance.



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Day research: smartwatches

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Battery Life Comparison Between Leading Smartwatch Brands

A comparison of battery life between leading smartwatch brands reveals significant differences. According to the guide on the 10 Best Smartwatches with Long Battery Life 2026 Models Tested, some brands exaggerate their battery life claims. The guide notes that the best battery life smartwatches do not work as expected, and actual performance may vary.

MIP (Memory-in-Pixel) displays, used by Garmin and COROS, are highlighted as a power-efficient technology, drawing power only when the pixels change. This suggests that smartwatches from these brands may have an advantage in terms of battery life. However, evidence on the exact battery life of each brand is thin, and further research is needed to determine the most accurate claims.

To find the perfect balance between smart features and lasting power, it is essential to research thoroughly and consider the specific needs of the user.

MIP Display Impact on Battery Life

Building on battery findings from prior research, MIP displays used by Garmin and COROS have been found to be power-efficient, drawing power only when pixels change. This is because every pixel has its own small memory cell and the pixel is only refreshed if a value changes. As a result, MIP displays are much more efficient on battery life, especially when compared to AMOLED displays. In fact, MIP watches like the FR955 don't need to turn the display off, unlike AMOLED watches like the FR965, as the screen only dims but doesn't turn off after a set timeout value.

The power efficiency of MIP displays is particularly noticeable in bright outdoor conditions, where they can remain visible without consuming significant battery power. In contrast, LCD and AMOLED displays need to shine brighter than the sun, resulting in higher battery consumption. However, it's worth noting that MIP displays may not be as bright as AMOLED displays, which can be a drawback in certain situations.

MIP vs. OLED/AMOLED/LCD: Power Consumption and Visuals

MIP (Memory-in-Pixel) displays, prominently featured in smartwatches from brands like Garmin and COROS, demonstrate significant power efficiency advantages, particularly for always-on functionality. Building on prior findings, MIP displays are inherently power-efficient because each pixel contains its own small memory cell, drawing power only when that pixel's value changes. This unique architecture allows MIP watches to keep their screens constantly on without incurring significant battery drain,

directly addressing the growing demand for longer battery life in smartwatches.

In contrast, OLED and AMOLED displays, while known for their vibrant colors and deep blacks, require a continuous power source for their light-emitting pixels to function. This fundamental difference means that maintaining an always-on display with OLED technology typically consumes more power than with MIP. Even with advancements like lower refresh rates, the energy lost through LTPS leakage can counteract potential power savings if pixels are not truly static, as constant updates demand more energy.

The key trade-off lies in visual quality. While the provided data doesn't explicitly detail MIP's visual characteristics, the nature of OLEDs as active light emitters implies they generally offer superior brightness, contrast, and color vibrancy compared to MIP displays. MIP screens often rely on ambient light for optimal visibility, making them excellent for outdoor use but potentially less striking in dimly lit environments. Therefore, smartwatches employing MIP displays prioritize multi-day battery life and always-on convenience over the vivid, high-fidelity visuals typically associated with OLED/AMOLED screens.

Sources:

Real-World Battery Life: MIP vs OLED Always-On Display Technology

Smartwatch battery life has become a critical differentiator in the wearables market, with display technology emerging as the primary factor influencing endurance. When comparing MIP (Memory-in-Pixel) displays used by brands like Garmin and COROS against the OLED panels found in Apple and Samsung watches, the architectural differences in how each technology powers its pixels create fundamentally different user experiences—especially when always-on display (AOD) is enabled.

The MIP Advantage

MIP displays achieve their exceptional power efficiency through a unique pixel memory architecture. Each individual pixel contains its own dedicated memory cell, allowing the screen to maintain an image without continuous power application. Unlike traditional displays that require constant refresh signals, MIP panels only consume meaningful power when a pixel's value actually changes. This means a watch face displaying the same time and complications for hours draws minimal current—essentially just enough to maintain the static image state.

This architectural choice makes MIP displays purpose-built for always-on functionality. Garmin and COROS have capitalized on this by marketing their watches as multi-week battery life devices, even with AOD active. The COROS VERTIX 2S exemplifies this approach, targeting ultra-endurance athletes who require absolute battery reliability in remote locations where charging access is impossible for days or weeks.

OLED Trade-offs

In contrast, OLED and AMOLED displays require each pixel to actively emit light or remain in a powered state to show any image. While OLED offers superior contrast ratios, true blacks through per-pixel dimming, and vibrant colors, maintaining an always-on watch face necessarily keeps the display's light-emitting diodes energized. Apple and Samsung have mitigated this through aggressive pixel cycling and lowered brightness strategies, but the fundamental physics remain: OLED pixels are either on or off, consuming power either way.

The Practical Gap

Real-world testing consistently shows this divide. Where MIP watches achieve 10-30+ days of AOD battery life, equivalent OLED watches typically manage 1-2 days. For users who prioritize functionality over visual flair—particularly outdoor athletes and minimalists—the MIP approach delivers tangible utility. However, OLED displays offer a richer interactive experience with smoother animations and superior readability in low-light conditions.

The choice ultimately depends on user priorities: MIP technology serves those who forget to charge their devices and value reliability over aesthetics, while OLED appeals to users who treat their smartwatch as a compact smartphone extension and charge daily.

Links to prior research:

- Prior findings established MIP pixel memory architecture fundamentals
- Confirmed Garmin/COROS as primary MIP implementers
- OLED power consumption characteristics documented from previous rounds

Sources:

Quantitative Always-On Battery Life: MIP vs OLED Smartwatch Displays

The fundamental architectural difference between display technologies creates a stark battery life divide in always-on smartwatch scenarios. MIP (Memory-in-Pixel) displays, employed by Garmin and COROS, achieve superior power efficiency by containing a memory cell within each pixel—a design that draws power only when pixel values actually change. OLED panels, used by Apple and Samsung watches, require continuous electrical current to maintain illumination even for static content, fundamentally undermining their always-on efficiency.

Brand-Specific Always-On Scenarios

Apple's Apple Watch employs an LTPO OLED panel that attempts to mitigate power draw through variable refresh rates, but the always-on mode still substantially reduces battery life compared to■■■■■■■■■■. Samsung's Galaxy Watch uses Super AMOLED with similar constraints—always-on display notably shortens runtime, typically requiring daily or near-daily charging.

The MIP architecture addresses a core inefficiency in OLED: while OLED pixels only consume power when lit (enabling true blacks to save power), the display controller must continuously refresh the panel to maintain static content. MIP displays eliminate this refresh requirement entirely for unchanging pixels, as each pixel's memory cell preserves its state without periodic refresh signals. This distinction becomes critical in always-on use, where the display shows relatively static information (time, date, widgets) for extended periods.

Sources:

- ## Real-World Battery Life: MIP vs OLED Flagship Smartwatches

MIP Watch Battery Performance

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independent reviews largely corroborate, with real-world testing showing only 10-15% variance from rated specifications.

OLED Watch Battery Performance

Apple

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Sources

ofzenandcomputing.com/best-smartwatches-with-long-batte...	https://www.ofzenandcomputing.com/best-smartwatches-with-long-battery-life/
gadgetschamp.com/smartwatches/best-battery-life-sm...	https://gadgetschamp.com/smartwatches/best-battery-life-smartwatches/
tech.yahoo.com/wearables/articles/mip-sick-battery-life-reaso...	https://tech.yahoo.com/wearables/articles/mip-sick-battery-life-reaso...
forums.garmin.com/sports-fitness/running-multisport...	https://forums.garmin.com/sports-fitness/running-multisport/f/forerun...
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